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Studierichting: Informatica
Oefeningenreeks 3B nr. 3

$$f(x) = \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

$$\frac{df}{dx}(x) = \frac{-1}{2} (x)^{-\frac{3}{2}}$$

$$\frac{d^2 f}{dx^2}(x) = \frac{-1}{2} \cdot \frac{-3}{2} (x)^{-\frac{5}{2}}$$

$$\frac{d^3 f}{dx^3}(x) = \frac{-1}{2} \cdot \frac{-3}{2} \cdot \frac{-5}{2} (x)^{-\frac{7}{2}}$$

$$\frac{d^4 f}{dx^4}(x) = \frac{-1}{2} \cdot \frac{-3}{2} \cdot \frac{-5}{2} \cdot \frac{-7}{2} (x)^{-\frac{9}{2}} = \frac{+105}{16\sqrt{x^9}} = \frac{105}{16x^4\sqrt{x}} = \frac{105\sqrt{x}}{16x^5}$$

References